SYSTEM DESIGN SPECIFICATION · VERSION 1.0

ResumeMatch PH

A Web-Based Résumé Builder and ATS-Inspired Job Matching System

Design documentation — wireframes, interface mockups, data model, matching algorithm, and interface specification.

STUDY

A Web-Based Résumé Builder Application for New Job Seekers in the Philippines

INSTITUTION

School of Computing, Holy Angel University

PILOT PARTNER

Royce Hotel & Casino, Clark Freeport Zone, Pampanga

IMPLEMENTATION

Angular 21 · Node.js / Express (TypeScript) · PostgreSQL (Supabase)

Prepared for adviser review. Sections marked **added** denote capabilities developed beyond the original proposal; these are reconciled in the accompanying Paper Revision Guide.

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01 System overview

The system joins two things normally sold as separate products: a guided résumé builder for people writing their first résumé, and an ATS-inspired matching engine that ranks those applicants for a partner company's HR department. Neither half is useful alone — the builder produces the structured data the engine needs, and the engine gives the builder a purpose beyond producing a PDF.

The two design commitments

ONE · THE DUAL-RÉSUMÉ MODEL

Every applicant has two résumés. The **structured résumé** is typed field by field into the wizard and is the only thing the engine scores — because the user entered it directly, nothing is lost to a parsing error. The **uploaded file** (PDF, Word, or image) is optional, is **never parsed and never scored**, and is passed to the recruiter to read with their own eyes. This is what allows the study to claim that no applicant is disadvantaged by document formatting.

TWO · NO AUTOMATIC REJECTION

The engine ranks; it never rejects. An applicant below the strong-match threshold remains visible to the recruiter and is additionally shown **related roles** reachable with their transferable skills. Every status change on an application is performed by a person and recorded together with the identity of who performed it.

Operating modes

MODE	STORAGE	PURPOSE
Demonstration	Browser local storage	No setup, functions without a network connection. Used for the defence demonstration and for usability testing where no server is available.
Server	PostgreSQL (Supabase)	The operational system: real accounts, persistence, email delivery, concurrent users.

Both modes implement a single interface contract, so every screen behaves identically in either mode and no page component knows which is active. Selecting a mode is a single configuration value.

02 Actors and permissions

Three roles. The third — the research team, acting as administrator — exists because recruiter accounts are *provisioned rather than self-served*: permitting open registration as an employer would allow any member of the public to obtain access to applicants' résumés.

ACTOR	ACCOUNT CREATION	PERMITTED	NOT PERMITTED
Job seeker	Self-registration, proven by a six-digit code sent to the address	Build and manage résumés, upload own file, view scores, apply, save openings, track and withdraw applications, export	View other applicants; view any posting's applicant list
Recruiter / HR	Requests access; an administrator approves, which creates the verified account	Manage company profile, post and close openings, view the ranked applicant list with score breakdowns, set statuses, message applicants	View another company's postings or applicants; reverse a withdrawal; self-register
Administrator added	Command line only — deliberately no web path exists	Approve or decline recruiter requests, assign department routing, view anonymous study metrics	Read any résumé or application content

03 Site map

Eighteen routes. Access control is enforced on the server as well as in the browser; the browser-side guard exists only to produce a clean redirect rather than an error.

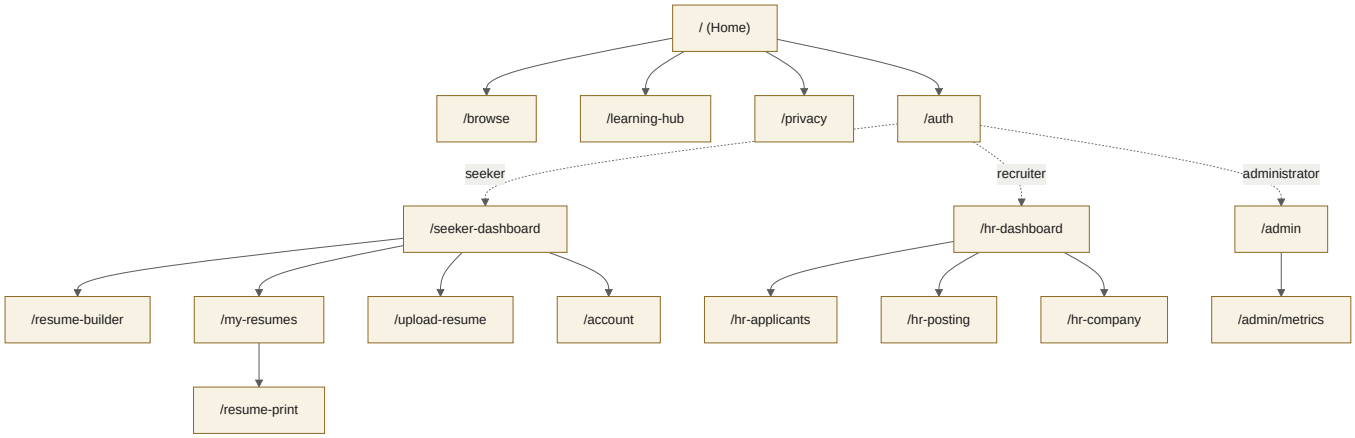


Figure 1. Site map. Solid lines are navigation; dotted lines are role-based redirection after sign-in.

ACCESS	ROUTES
Public	Home, openings, learning hub, privacy notice, sign in / register, not-found
Job seeker	Dashboard, résumé builder, upload and extract, my résumés, print view, account
Recruiter	Dashboard, applicants, post an opening, company profile, account
Administrator	Recruiter requests, study metrics, account

04 User flows

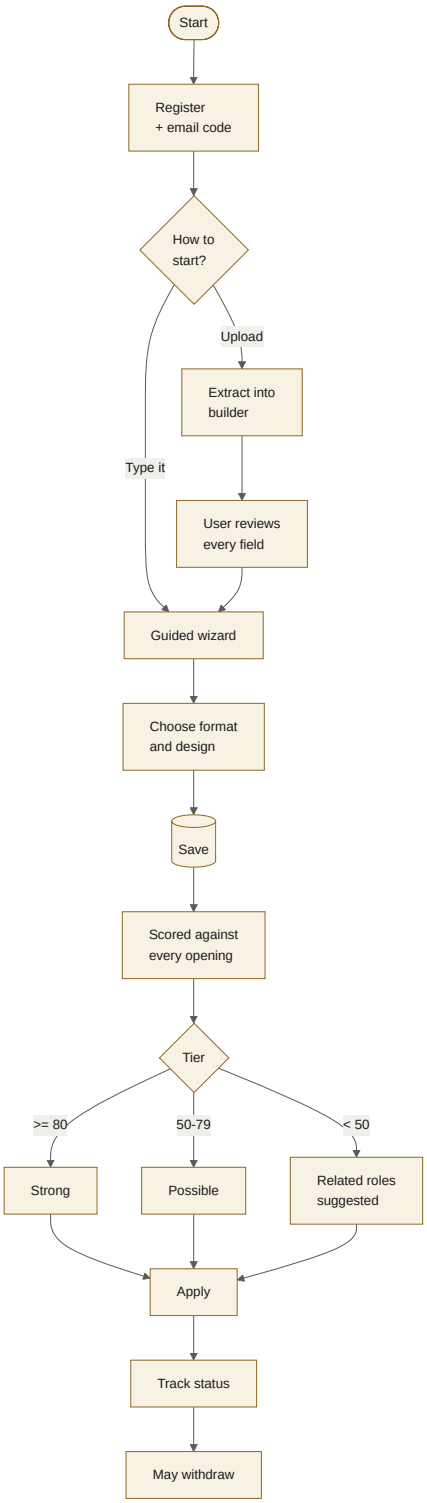


Figure 2. Job seeker flow, from registration to a tracked application.

> The **Related** branch is the study's central contribution: a low score never terminates the flow. The applicant may still apply, and is simultaneously directed toward roles their existing skills do fit.

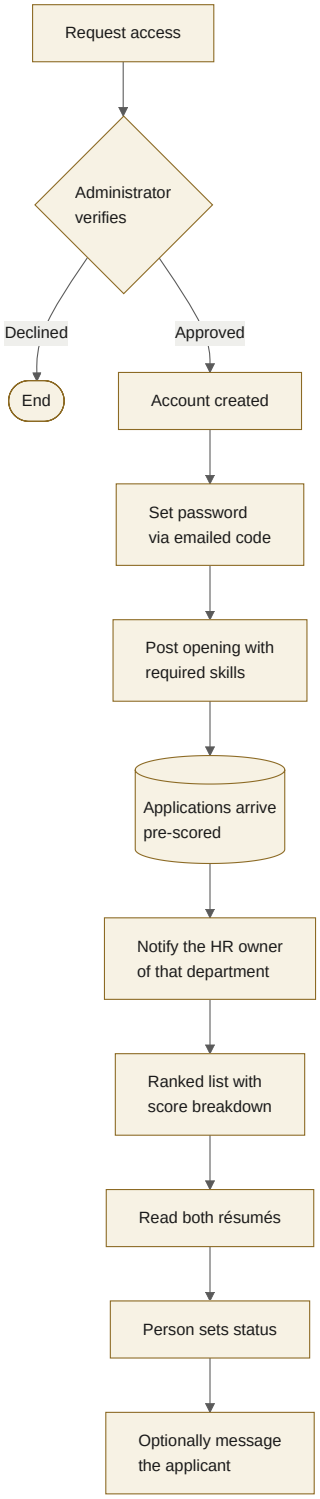


Figure 3. Recruiter flow, from access request to a hiring decision.

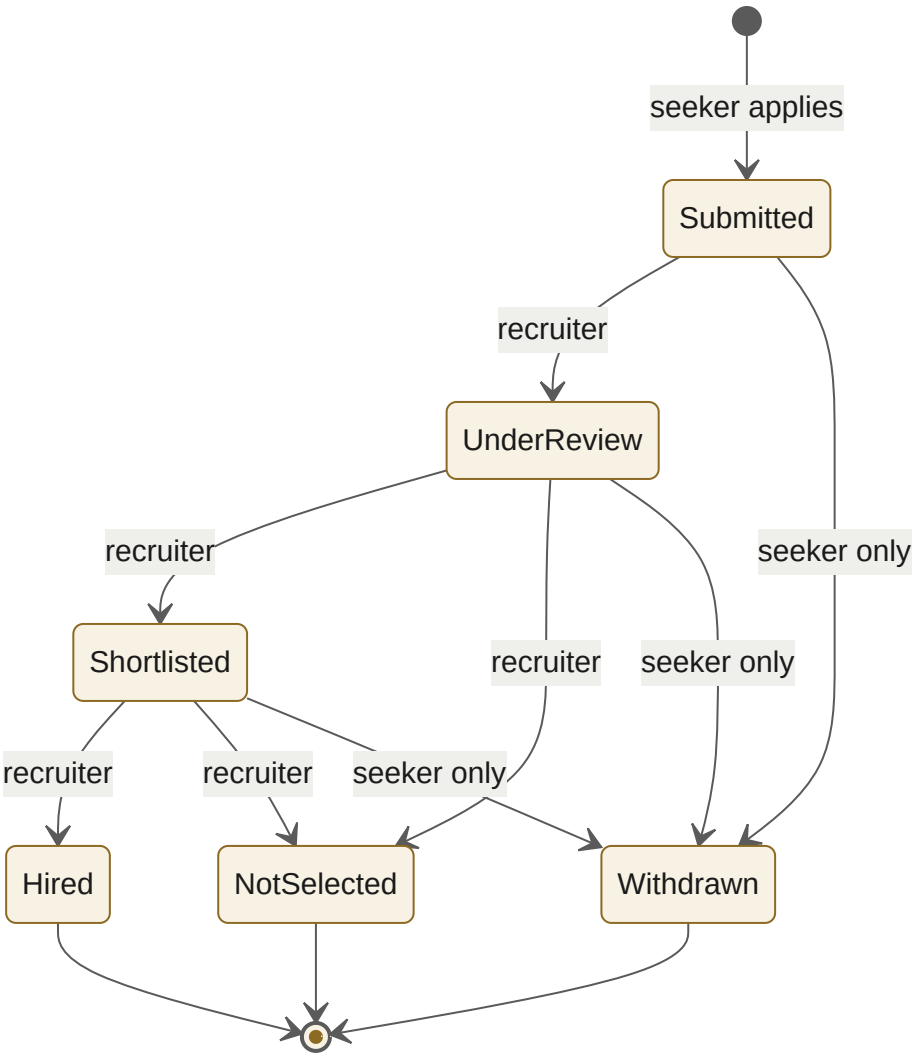


Figure 4. Application status lifecycle. Every transition is performed by a person.

added **Withdrawn** is set by the applicant alone. A recruiter can neither set it nor overrule it, and it is unavailable once a final decision exists — there is nothing remaining to withdraw from.

05 Wireframes

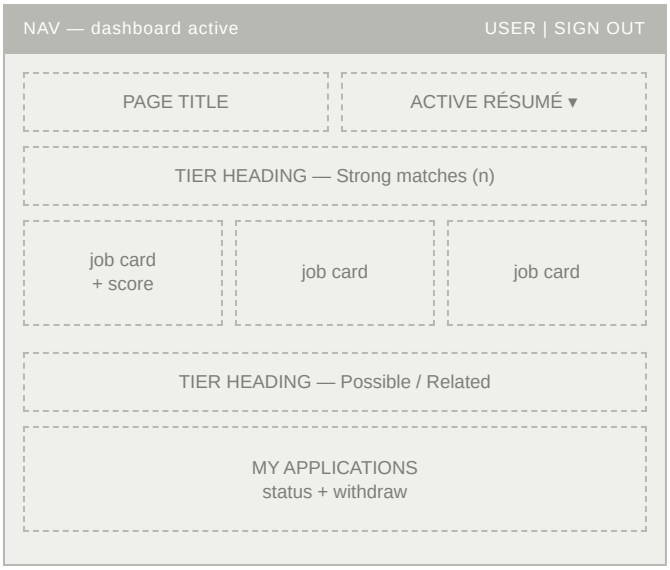
Low-fidelity layouts showing structure and hierarchy only, deliberately without colour or final wording, so that layout decisions can be reviewed before any visual design is agreed.

W-01 · Home (public)



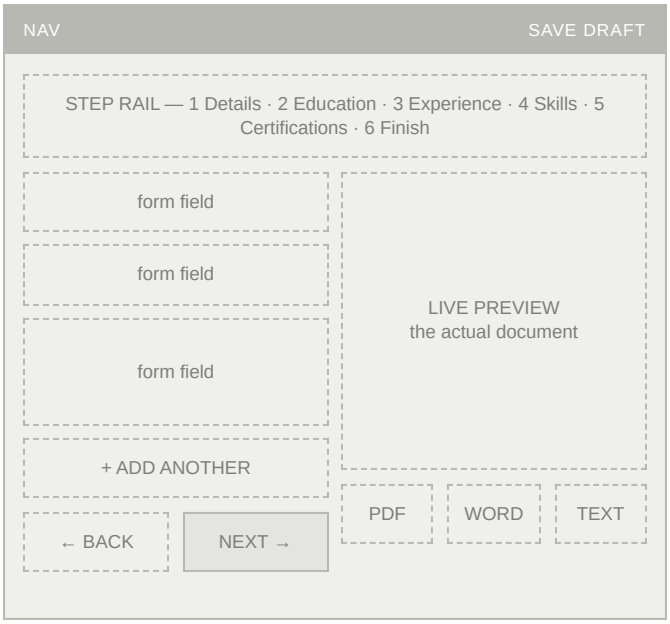
The product preview sits above the fold: the argument is made visually — three applicants, none rejected — rather than in prose.

W-02 · Job seeker dashboard

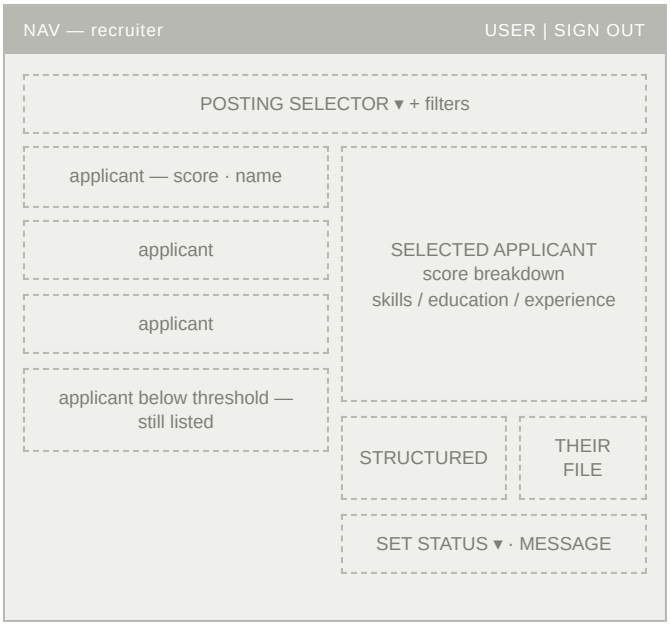


Results are grouped by tier rather than presented as one ranked list, so that "related" reads as a category of opportunity rather than as failure.

W-03 · Résumé builder (two-pane)



W-04 · Recruiter — ranked applicants



The preview is the real document rather than a thumbnail, and updates on every keystroke, so the user is never uncertain what they are producing. It collapses to a single column on narrow screens.

Both résumés are reachable side by side and the score breakdown is always shown, so the recruiter sees why a figure was produced rather than only the figure.

W-05 · Dashboard at 390 pixels (mobile)

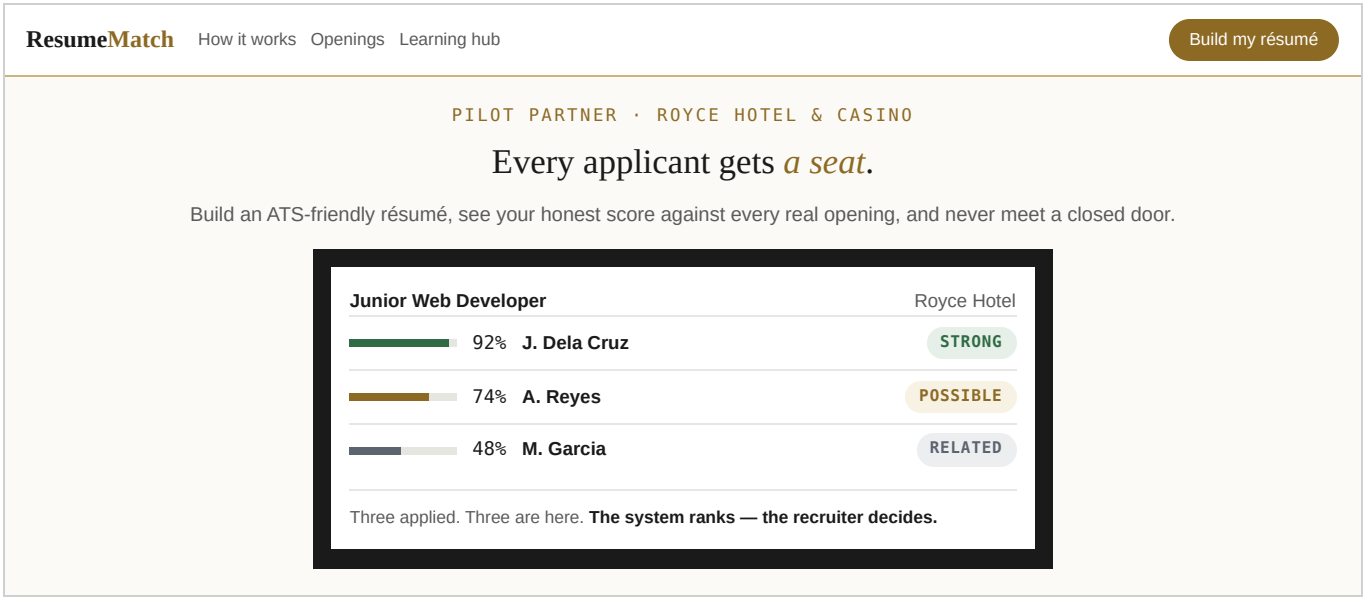


Every multi-column grid collapses to a single column, navigation becomes a menu button, and no page scrolls horizontally at any width.

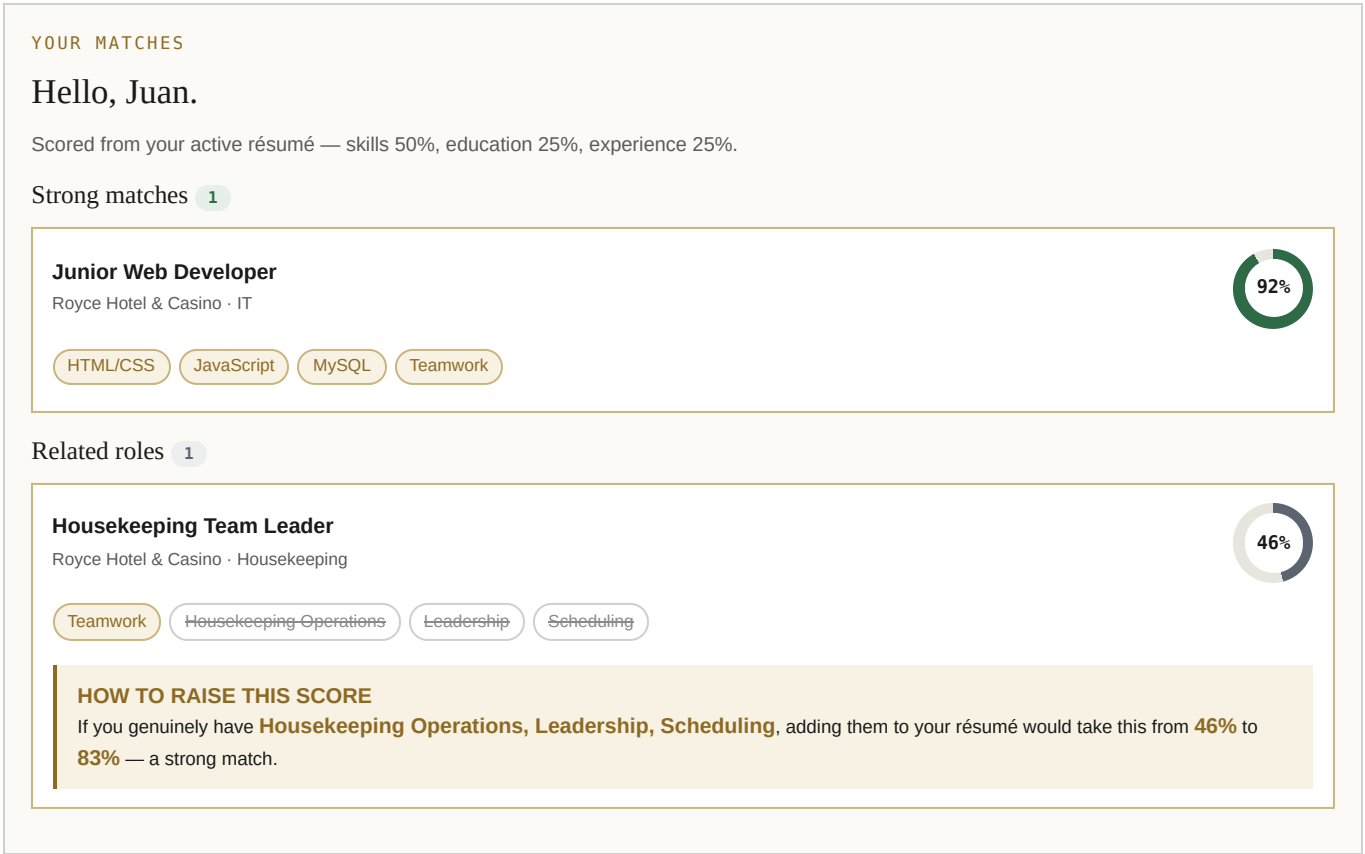
06 Interface mockups

The same screens with the visual design applied. These correspond to the system as built.

M-01 · Home — hero and product preview



M-02 · Job seeker dashboard — tiered matches and score guidance



> Struck-through chips are the unmet requirements; the highlighted panel states the score that would result were they present. Together they make a low score actionable rather than final.

The eight ATS-safe résumé designs

#	DESIGN	HEADER TREATMENT	SECTION TREATMENT	TYPEFACE
1	Classic	Centred	Ruled, centred	Serif
2	Modern	Left, accent rule beneath	Accent bar at left	Sans
3	Minimal	Left, light weight	Hairline rule	Sans (light)
4	Executive	Left, with target role	Double rule	Serif
5	Banner	Filled dark band	Ruled	Sans
6	Split	Name left, contact right	Centred with marks	Serif
7	Bar	Left, compact	Filled heading bands	Sans
8	Categorized	Left, with accent rail	Skills grouped by category	Sans

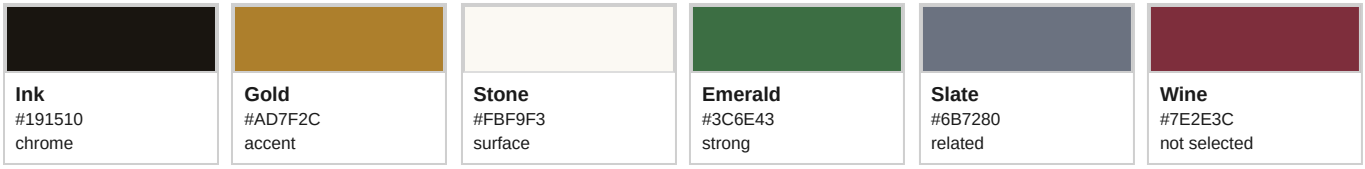
> **All eight are ATS-safe by construction:** a single-column layout so that a parser reads one continuous flow, standard section headings the parser searches for by name, real selectable text in standard typefaces, and no tables, text boxes, icons, or images. The design alters only appearance; the words, and therefore the compatibility score, are identical.

Format and design are separate choices. The *format* — chronological, functional, or combination — determines the order in which sections appear. The *design* determines appearance. Three formats and eight designs yield twenty-four combinations.

07 Design system

An identity derived from hospitality, drawn from the pilot partner's own brand language: dark chrome, burnished gold, and warm paper surfaces.

COLOUR



Ink	Gold	Stone	Emerald	Slate	Wine
#191510	#AD7F2C	#FBF9F3	#3C6E43	#6B7280	#7E2E3C
chrome	accent	surface	strong	related	not selected

> **Semantic discipline.** Exactly one colour is assigned to each match tier, and "related" is rendered in **slate — a neutral, not a warning colour**. Rendering it in red or amber would contradict the system's central claim that a low score is a redirection rather than a failure.

TYPOGRAPHY

ROLE	TYPEFACE	APPLIED TO
Display	Fraunces (serif)	Headings and the wordmark; carries the hospitality tone
Body	Manrope (sans-serif)	All running text, forms, and controls
Label	JetBrains Mono	Section labels, scores, statuses — anything numeric or systematic

PRINCIPAL COMPONENTS

COMPONENT	BEHAVIOUR
Score ring	A circular dial coloured by tier. It never empties completely: even a low score retains an arc, reinforcing visually that the applicant was redirected rather than rejected.
Skill chip	Matched requirements are filled; unmet ones are struck through and muted, so the gap is legible at a glance.
Tier badge	Strong, Possible, or Related, using the semantic colours above.
Notification	A transient confirmation. Every instance is a live region, so users of assistive technology receive the same confirmation as sighted users.

ACCESSIBILITY AND THEMING ADDED

- Light and dark appearance on every page, remembered per browser and honouring the operating-system preference on first visit.
- Keyboard-reachable controls with a visible focus indicator, and a skip-to-content link.
- All status confirmations announced to assistive technology.
- Responsive down to 390 pixels with no horizontal scrolling; reduced-motion preferences respected.

08 Use case diagram

Revised to include the administrator, who was absent from the original two-actor diagram.

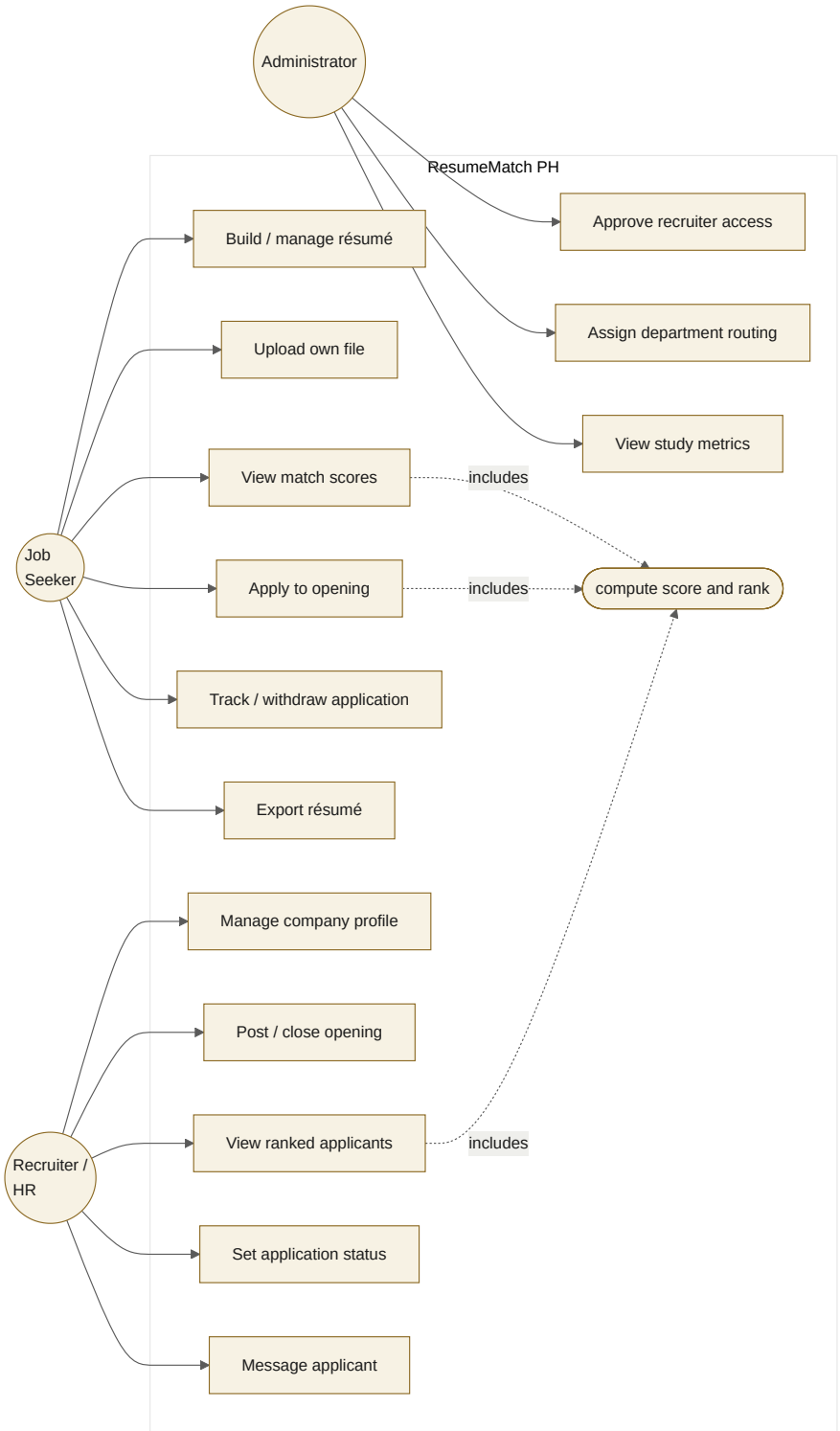


Figure 5. Use case diagram, revised to three actors. Scoring and ranking is an included behaviour rather than an actor-initiated case.

09 Database design

PostgreSQL, hosted on Supabase. Seventeen tables; the diagram below shows the core entities and their relationships.

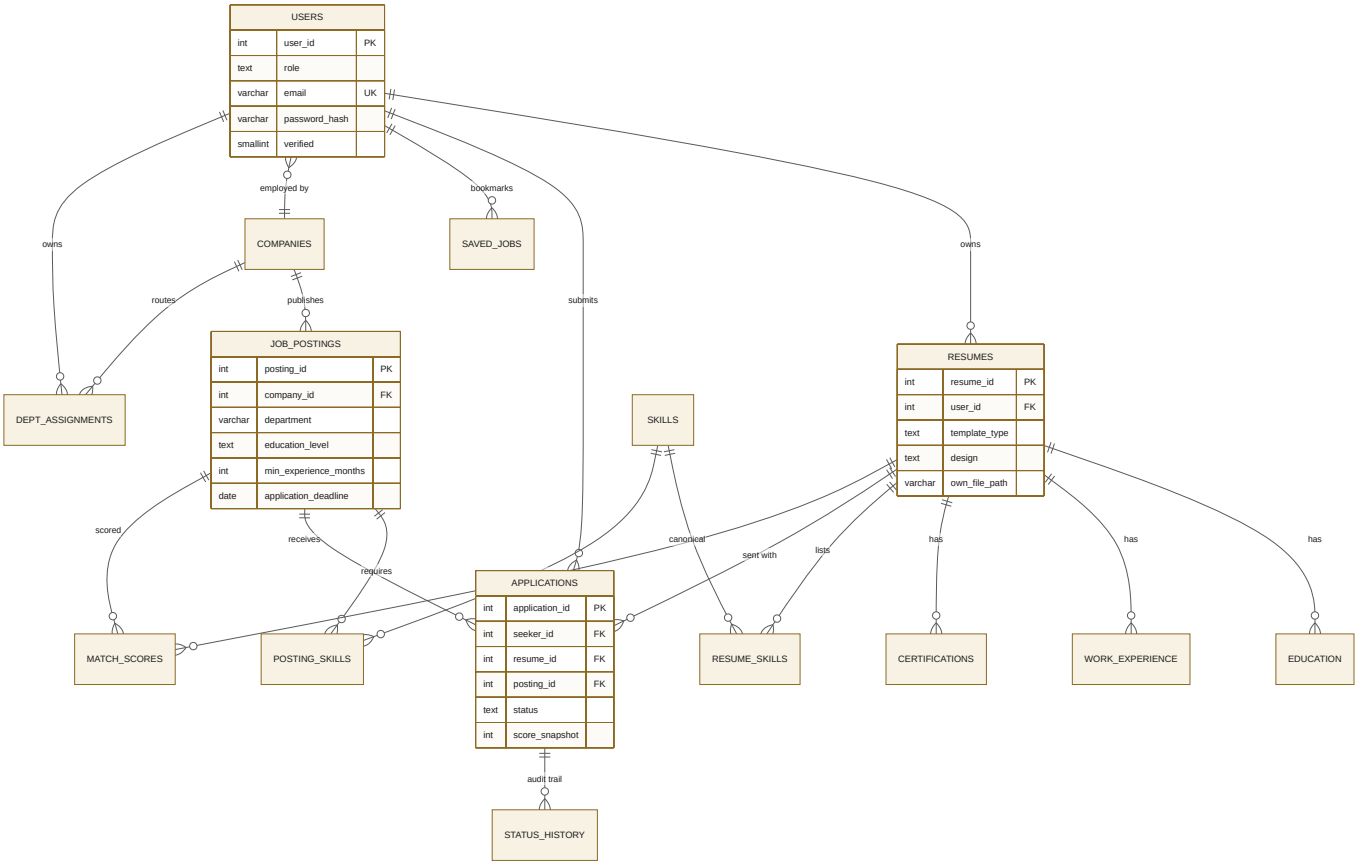


Figure 6. Entity relationship diagram (core entities).

Design decisions

DECISION	RATIONALE
A canonical skills table	Every skill string resolves to a single row, so a skill named on a résumé and the same skill named on a posting are literally the same record. The comparison is therefore an identifier match rather than a text match.
Score stored on the application	The score is recorded as it stood at the moment of applying. If the applicant later edits their résumé, the historical record of what the recruiter actually saw is preserved — without which the study's data would not be interpretable.
A status history table	Every status change records the identity of the person who made it. This is the direct evidence for the claim that decisions are made by people and not by the system.
File paths, never file contents	An uploaded document is stored outside any publicly reachable directory and served only through an endpoint that checks who is asking. It is never read by the matching engine.

10 Matching algorithm

Rule-based and fully published — no machine learning participates. An applicant can reproduce their own score by hand, which is the point of publishing it.

$$score = (S \times 0.50 + E \times 0.25 + X \times 0.25) \times 100$$

rounded half-up to a whole percentage

TERM	COMPONENT	WEIGHT	DEFINITION
S	Skills	50%	Matched required skills divided by total required skills. A posting stating no required skills yields full credit.
E	Education	25%	The applicant's highest attainment divided by the level required, on a scale of zero to five, capped at one. **Partial credit, never zero**: an applicant one level short still earns proportional credit.
X	Experience	25%	Total months of experience divided by the months required, capped at one. No stated requirement yields full credit.

TIERS

Strong — 80 and above **Possible** — 50 to 79 **Related** — below 50

An applicant in the Related tier is shown alternative roles reachable with their transferable skills, remains fully visible to the recruiter, and may still apply.

SKILL COMPARISON REFINED

Skills are compared by canonical form rather than literal text: capitalisation and punctuation are normalised, and a defined set of synonyms is folded together, so that "JS" and "JavaScript", or "MS Office" and "Microsoft Office", are treated as the same skill. Without this step an applicant would be penalised for vocabulary alone, producing precisely the kind of false negative the study argues against. The synonym set is deliberately conservative: it unifies terms denoting the same skill and never relates merely similar ones, so no applicant is credited with a qualification they did not claim.

PROPERTIES THAT FOLLOW FROM THE DESIGN

- **Deterministic.** Identical inputs produce an identical score on every occasion.
- **Transparent.** The weights are published within the interface, and the applicant is shown the same breakdown the recruiter sees.
- **Adjustable without redeployment.** Weights and thresholds are held in a settings table, supporting the commitment to refine them iteratively during testing.
- **Implemented twice and proven identical.** An authoritative server implementation and a browser implementation for instant preview are verified equal across five hundred generated cases.

11 Interface specification

All operations are addressed as `POST /api/<resource>/<action>` and return either a success object or a human-readable error; they never raise an exception at the caller. The session is carried in a signed, HTTP-only cookie.

OPERATION	ROLE	PURPOSE
auth/login · logout · me	any	Session lifecycle. Sign-in responds identically whether or not the account exists.
auth/send-code · register · reset-password	public	Email verification by six-digit code, registration, password reset.
auth/change-email-start · confirm	any	Email change, proven by a code sent to the new address.
auth/export · delete-account	any	Data portability and erasure (Republic Act 10173).
resumes/list · get · save · delete · duplicate	seeker	Résumé management. Ownership is enforced on the server.
resumes/upload · remove-file · file	seeker, HR	The applicant's own document, served only through an access check.
analyze/resume	seeker	Optional extraction of an uploaded document into builder fields. Never scores.
postings/list · mine · save · toggle	public, HR	Job openings. Postings past their deadline close themselves.
match/score · recommendations	seeker	Score a single pairing; return all openings grouped by tier.
applications/apply · mine · withdraw	seeker	Apply, track, withdraw. Duplicate applications are refused.
applications/ranked · set-status · invite	recruiter	Ranked list, status decision, message to an applicant.
saved/list · toggle	seeker	Bookmarked openings.
admin/requests · approve · decline	administrator	Recruiter provisioning.
admin/dept-assignments · assign-department	administrator	Departmental routing of applications.
metrics/summary	administrator	Anonymous aggregates for the study.

12 Testing strategy

Automated tests provide evidence for the ISO 25010 **functionality** and **reliability** characteristics; user acceptance testing addresses **usability**.

SUITE	TESTS	WHAT IT ESTABLISHES
Matching engine	29	That the published weights and tier thresholds behave exactly as documented; that each scoring component follows its stated rule; that the tier boundaries are exact; that skill synonyms behave and remain conservative; and, through a five-hundred-case sweep, that the server and browser implementations are identical.
Application interface	16	Authentication — including that sign-in cannot be used to discover whether an account exists — role-based access control, data ownership, prevention of duplicate applications, the withdrawal rules, company scoping, and the rule that an uploaded file is never involved in scoring.
User acceptance	—	ISO 25010 usability, functionality, and reliability, evaluated by job seeker and HR respondents.

Continuous integration. Every change builds the application, checks types, and executes both suites against a temporary database. A modification that breaks the published formula therefore cannot be merged unnoticed.

13 Build sequence

The order in which the system was constructed, and the condition that marks each stage complete.

STAGE	WORK	COMPLETE WHEN
1	Database schema	A single SQL file creates every table and seeds demonstration data, and may be re-run safely.
2	Design system	Colour, type, and component definitions exist once and every page inherits them.
3	Accounts and roles	Registration with email verification, sign-in, and access control enforced on the server.
4	Résumé builder	Six-step wizard with live preview, saving, editing, duplication, and export.
5	Matching engine	Scores computed identically on server and browser, covered by automated tests.
6	Postings and applications	A recruiter posts; an applicant applies; the ranked list shows the breakdown.
7	Administration	Recruiter approval, departmental routing, and study metrics.
8	Refinement	Responsive to 390 pixels, accessible, dark appearance, and an onboarding tour.

RUNNING THE SYSTEM

Demonstration mode — no database, no keys, functions offline:

```
cd client  →  npm install  →  npm start  →  localhost:4200
```

Server mode — import the SQL file into Supabase, set the database connection string, start the interface server, and switch the client configuration to server mode.

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